Cost-benefit analysis

Uncertainty and risk

Overview

- Decision tree and expected NPV
- Sensitivity, scenario analysis, and simulation
- Quasi-option value

Uncertainty

- Consider a municipality that faces increasing flooding and occasional damaging storm surges.

- The municipal council must decide whether to invest in

  - a permanent coastal flood barrier now,

  - delay while commissioning better climate-risk projections, or

  - build a smaller interim protection.

- Future storm frequency and severity are **uncertain** — and that <u>uncertainty drives whether the project is welfare-improving</u>.

Key points

- Future costs and benefits are uncertain → CBA must explicitly model that uncertainty.

- Common tools:
  To get an understanding of the risk
  —Expected NPV, sensitivity/scenario analysis, simulation.
  When information might mitigate uncertainty
  —quasi-option value.

- Uncertainty can change a “profitable” project into an unprofitable one (and vice versa).

- Important policy choices: accept now, reject, or delay (value of information / option to wait).

Modeling uncertainty with **contingencies** (one-period ENPV)

- Represent future uncertainty as a small, tractable set of **exhaustive & mutually-exclusive** contingencies (states) and attach probabilities to them — then evaluate the project by the probability-weighted net benefit.

- Choose a representative set of states (e.g. **High / Moderate / Low rainfall**) that span plausible outcomes.

- For each state k, specify benefits B_k and costs C_k (or net benefit $NB_k = B_k - C_k$).

- Assign probabilities p_k (objective: historical frequencies; or subjective: expert/forecast).

- Compute expected net benefit (one period):

$$\text{ENPV} = \sum_{k=1}^{K} p_k \,(B_k - C_k) = \sum_{k=1}^{K} p_k \, NB_k, \qquad \sum_{k} p_k = 1.$$

- Decision rule (simple): **Accept if ENPV > 0**; otherwise reject or consider delay/staging.

High sensitivity to probability assignment

- Up-front public investment: $I_0 = 300$.
- If built, the barrier **avoids** the storm damage amounts below (assume the barrier prevents the listed damages):
  - Extreme storm damage avoided: $D_E = 1200$.
  - Moderate storm damage avoided: $D_M = 250$.
  - Low/no-storm damage avoided: $D_L = 20$.

Baseline priors: $p_E = 0.04,\ p_M = 0.16,\ p_L = 0.80$

$$\begin{aligned}\text{Expected avoided damage} &= 0.04 \times 1200 + 0.16 \times 250 + 0.80 \times 20 \\ &= 48 + 40 + 16 = 104 \\ ENPV &= -300 + 104 = -196 \quad \text{(Do not build)}\end{aligned}$$

Updated priors (moderate risk): $p_E = 0.12,\ p_M = 0.28,\ p_L = 0.60$

$$\begin{aligned}\text{Expected avoided damage} &= 0.12 \times 1200 + 0.28 \times 250 + 0.60 \times 20 = 226 \\ ENPV &= -300 + 226 = -74 \quad \text{(Still negative, much closer)}\end{aligned}$$

High-end projection (worse risk): $p_E = 0.20,\ p_M = 0.30,\ p_L = 0.50$

$$\begin{aligned}\text{Expected avoided damage} &= 0.20 \times 1200 + 0.30 \times 250 + 0.50 \times 20 = 325 \\ ENPV &= -300 + 325 = +25 \quad \text{(Build)}\end{aligned}$$

High sensitivity to probability assignment

- Small-to-moderate changes in estimated probabilities of extreme storms (driven by climate projections, local subsidence, or new modelling) can **flip** the ENPV sign and thus reverse the social decision.

- For a publicly funded safety project, the council must therefore
(i) quantify uncertainty explicitly and
(ii) consider the value of improved information or staged/adaptive investments.

Multi-period ENPV

- Multi-period ENPV:

$$\text{ENPV} = -I_0 + \sum_{t=1}^{T} \frac{\text{ENPV}_t}{(1+i)^t},$$

  where ENPV_t is the expected net benefit in period t and i is the discount rate.

- Build the project's expected net benefits year-by-year, discount them, and subtract the initial investment.

- Practically we often treat these expected values as if they were the "true" outcomes — acceptable when risk is pooled across many projects or many affected people (realized outcomes average out).

- However, treating expectations as certainties is conceptually wrong for individual WTP under uncertainty — always better to report distributions or probabilities when stakes are large.

Multi-period ENPV

- The formula assumes period-by-period contingencies (and their probabilities) are **independent of earlier actions**.
- **Counterexample:** a vaccination program — choosing to vaccinate in $t = 1$ changes infection probabilities in later periods (path dependence).
    - When actions affect future risk, use a decision-tree / dynamic programming / Markov model (or explicitly model learning and feedback) instead of the simple sum-of-expectations.

Decision tree and ENPV

- What a decision tree does and how to solve it
—It turns a sequential CBA with timing and contingent events into a solvable diagram
(decisions = squares, chance/nature = circles).

- Draw the tree left → right: initial decision → chance nodes → later decisions/outcomes.

- Solve by **backward induction**: replace terminal chance nodes with their expected payoffs and prune dominated branches (choose the lower expected cost / higher expected net benefit).

- Use when later probabilities or payoffs depend on earlier actions (i.e., path-dependence) — otherwise simple year-by-year ENPV suffices.

Decision tree and ENPV

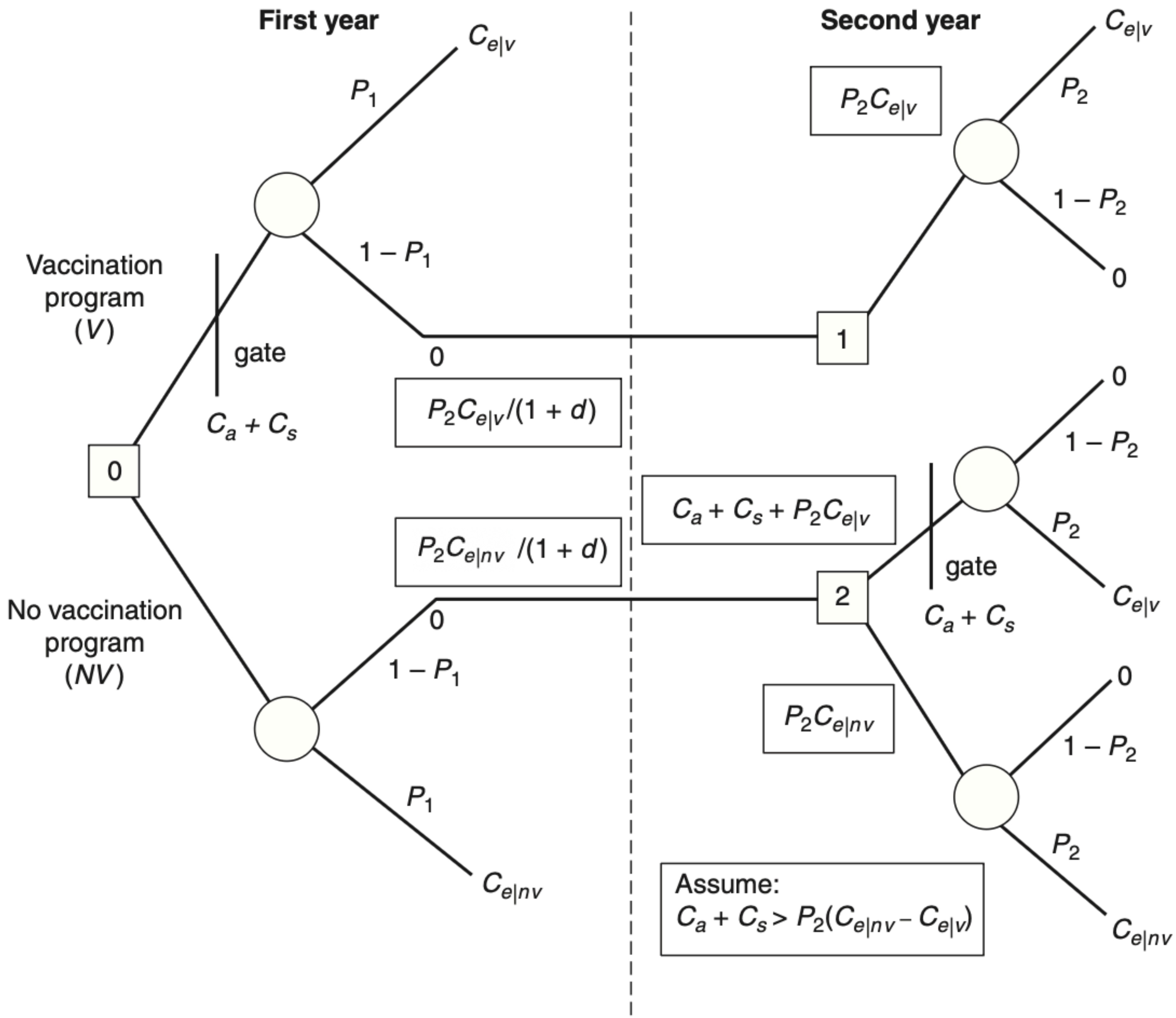


Figure 11.2 Decision tree for vaccination program analysis.

Decision tree and ENPV

Setup / notation

- Three cost items: C_a (admin cost), C_s (side-effect cost), $C_e|_v, C_e|_{nv}$ (epidemic cost with / without vaccination).
- Probabilities: P_1 (epidemic in period 1), P_2 (if no epidemic in period 1, probability in period 2). Discount rate d.

Node-2 (late decision):

- If *vaccinate* at node 2: expected cost $= C_a + C_s + P_2 C_e|_v$.
- If *not vaccinate*: expected cost $= P_2 C_e|_{nv}$.
- If $C_a + C_s > P_2\big(C_e|_{nv} - C_e|_v\big)$ then **vaccinate at node 2 is dominated** → choose *not vaccinate* at node 2.

Decision tree and ENPV

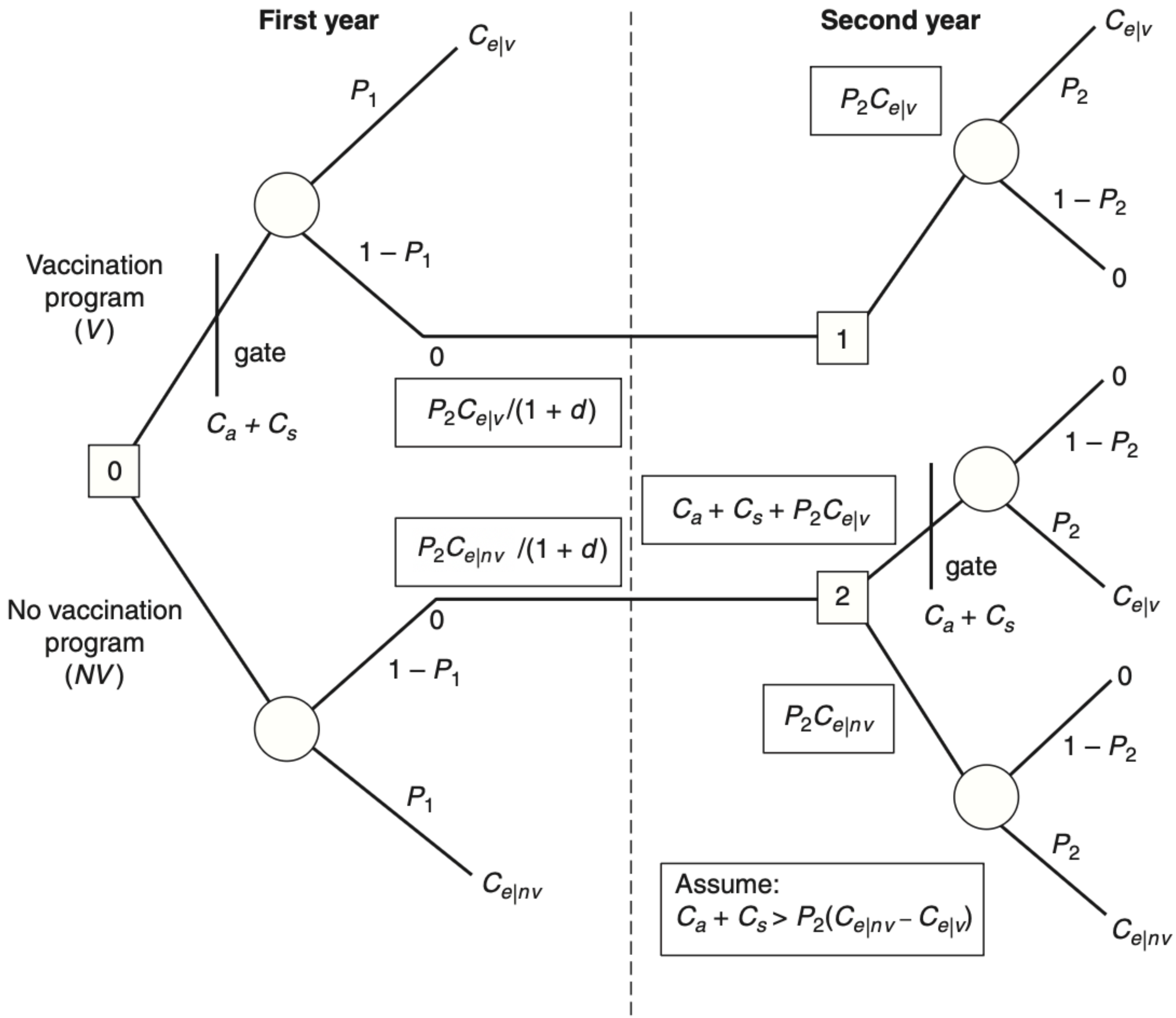


Figure 11.2 Decision tree for vaccination program analysis.

Decision tree and ENPV

Node-0 (overall expected cost)

$$\begin{aligned} EC(V) \ &= C_a + C_s + P_1 C_e|_v \ + \ (1 - P_1)\frac{P_2 C_e|_v}{1+d}, \\ EC(NV) \ &= P_1 C_e|_{nv} \ + \ (1 - P_1)\frac{P_2 C_e|_{nv}}{1+d}. \end{aligned}$$

Net benefit of vaccinating now (ENPV) = $EC(NV) - EC(V)$.

Numeric example (expressed as multiples of $C_{e|nv}$)

Assume: $P_1 = 0.4, \ P_2 = 0.2, \ d = 0.05, \ C_{e|v} = 0.5C_{e|nv}, \ C_a = 0.1C_{e|nv}, \ C_s = 0.01C_{e|nv}$.

Substitute →

$$\begin{aligned} EC(V) &= 0.367142857 \ C_{e|nv}, \\ EC(NV) &= 0.514285714 \ C_{e|nv}, \\ ENPV &= EC(NV) - EC(V) = 0.147142857 \ C_{e|nv} > 0. \end{aligned}$$

Backward induction: *do not* initiate at node 2 (dominated); compare at node 0 and **vaccinate now** because ENPV $= 0.147\, C_{e|nv} > 0$ (vaccination reduces expected social cost).

Sensitivity analysis

- Find how **robust** the base-case net benefits are to changes in key assumptions; detect parameters that can *flip* the decision.
- When to use: when estimates depend on many uncertain inputs or when stakes are large.
- Three practical forms:
  - **Partial sensitivity** — vary one parameter at a time, hold others fixed (good for identifying breakeven values).
  - **Worst / best-case analysis** — combine most pessimistic / optimistic plausible values to bound outcomes.
  - **Monte Carlo (global) simulation** — treat key inputs as random variables and compute the distribution of net benefits (used when many parameters matter).

Sensitivity analysis

- Build **parameters**–based model:

Variable	Value (millions of dollars)	Formula
C_a	0.404	$o + (V_h + V_l)q$
C_s	9.916	$\alpha(V_h + V_l)(wt + m_h L)$
$C_{e\mid nv}$	147.3	$i[rN(wt + m_h L) + (1 - r)N(wt + m_t L]$
$C_{e\mid v}$	87.9	$(\mathrm{i} - \theta ve)\{(rN - eV_h)(wt + m_h L) + [(1 - r)N - eV_l](wt + m_l L)\}$
ECv	55.7	$C_a + C_s + p_1 C_{e\mid v} + (1 - p_1)\, p_2\, C_{e\mid v}/(1 + d)$
EC$_{nv}$	76.0	$p_1 C_{e\mid v} + (1 - p_1)p_2 C_{e\mid v}/(1 + d)$
$E[NB]$	20.3	$\mathrm{EC}_{nv} - \mathrm{EC}_v$

Sensitivity analysis

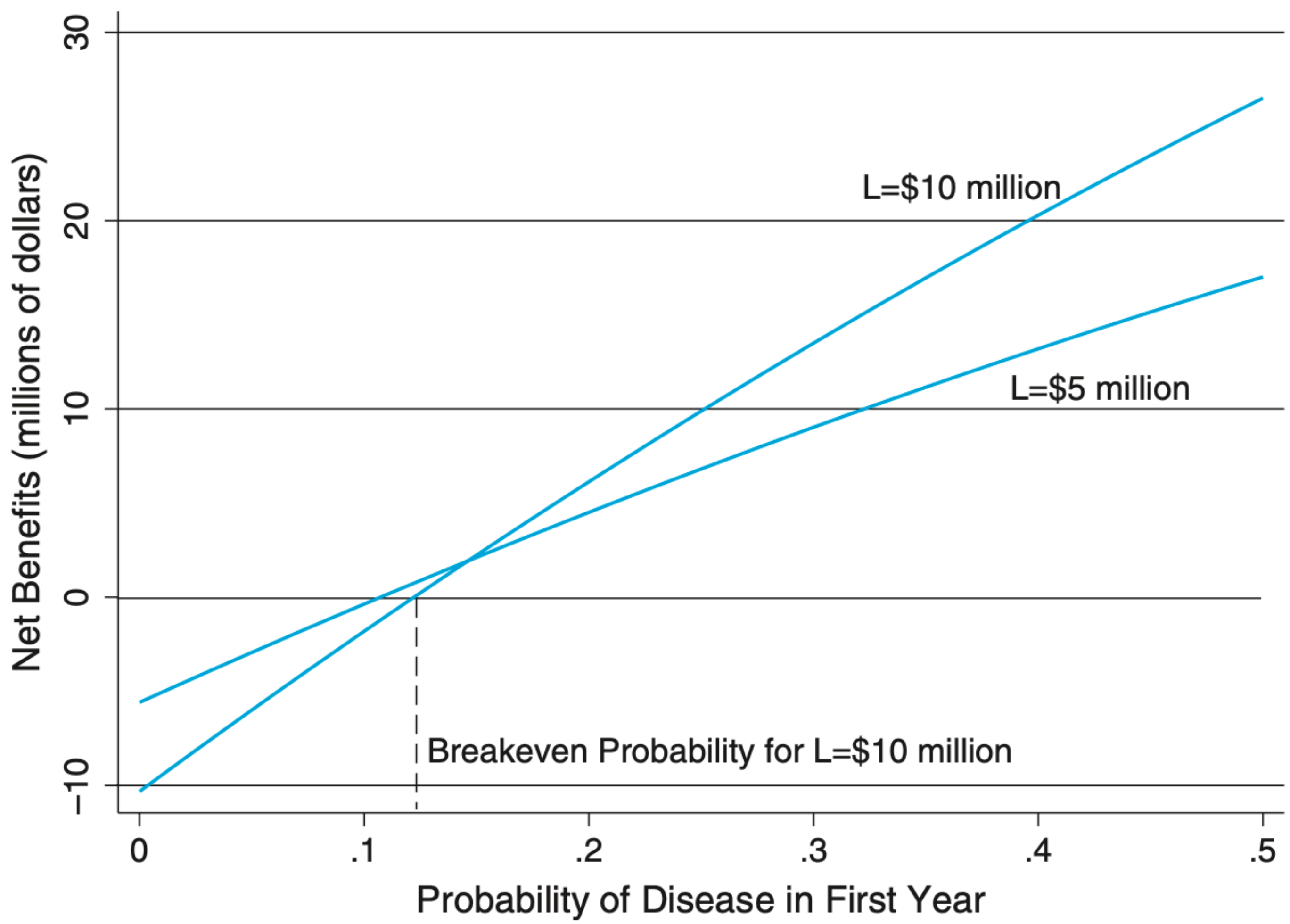
- Allow **parameters** to move across a range of values

Table 11.2 *Base-Case Values for Vaccination Program CBA*

Parameter	Value [range]	Comments
County population (N)	380,000	Total population in the county
Fraction high risk (r)	.06 [.04, .08]	One-half population over age 64
Low-risk vaccination rate (v_l)	.05 [.03, .07]	Fraction of low-risk persons vaccinated
High-risk vaccination rate (v_h)	.60 [.40, .80]	Fraction of high-risk persons vaccinated
Adverse reaction rate (α)	.03 [.01, .05]	Fraction vaccinated who become high-risk
Low-risk mortality rate (m_l) [.000025, .000075]	.00005	Mortality rate for low-risk infected
High-risk mortality rate (m_h) [.0005, .00015]	.001	Mortality rate for high-risk infected
Herd immunity effect (θ)	1.0 [.5, 1.0]	Fraction of effectively vaccinated who contribute to herd immunity effect
Vaccine effectiveness rate (e)	.75 [.65, .85]	Fraction of vaccinated who develop
Hours lost (t)	24 [18, 30]	Average number of work hours lost to illness
Infection rate (i)	.25 [.20, .30]	Infection rate without vaccine
First-year epidemic probability (p_1)	.40	Chance of epidemic in current year
Second-year epidemic probability (p_2)	.20	Chance of epidemic next year
Vaccine dose price (q)	\$9/dose	Price per dose of vaccine
Overhead cost (o)	\$120,000	Costs not dependent on number vaccinated
Opportunity cost of time (w)	\$20/hour	Average wage rate (including benefits) in the county
Value of life (L)	\$10,000,000	Assumed value of life
Discount rate (d)	.035	Real discount rate
Number high-risk vaccinations (V_h)	13,680	High-risk persons vaccinated: $v_h rN$
Number low-risk vaccinations (V_l)	17,860	Low-risk persons vaccinated: $v_l(1-r)N$
Fraction vaccinated (v)	.083	Fraction of total population vaccinated: $rv_h + v_l(1-r)$

Partial sensitivity analysis

- Plot net benefits vs the parameter of interest (e.g. epidemic probability p1).
- Show base case & alternate assumption lines (e.g. different values of value-of-life).
- Highlight the **breakeven** point (value of the parameter where NB = 0) — this communicates the decision threshold.

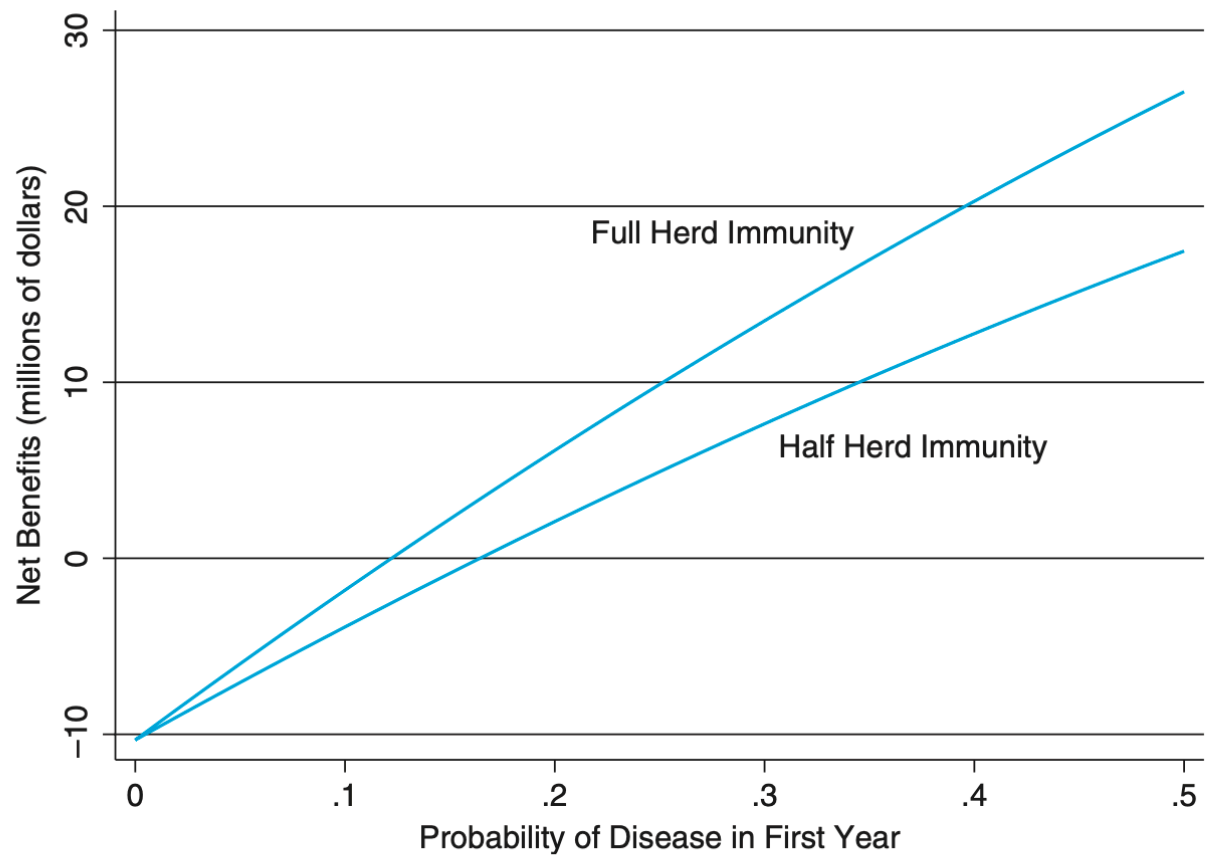


Partial sensitivity analysis

- Run the same partial sensitivity for two plausible model variants (e.g. full vs half herd immunity). Plot both curves on one chart to show how model structure matters.

- Sensitivity is not only to parameter values but also to structural assumptions (model form, interaction effects).

- Worst/best case: compute NB using the least / most favorable plausible values for each key parameter.



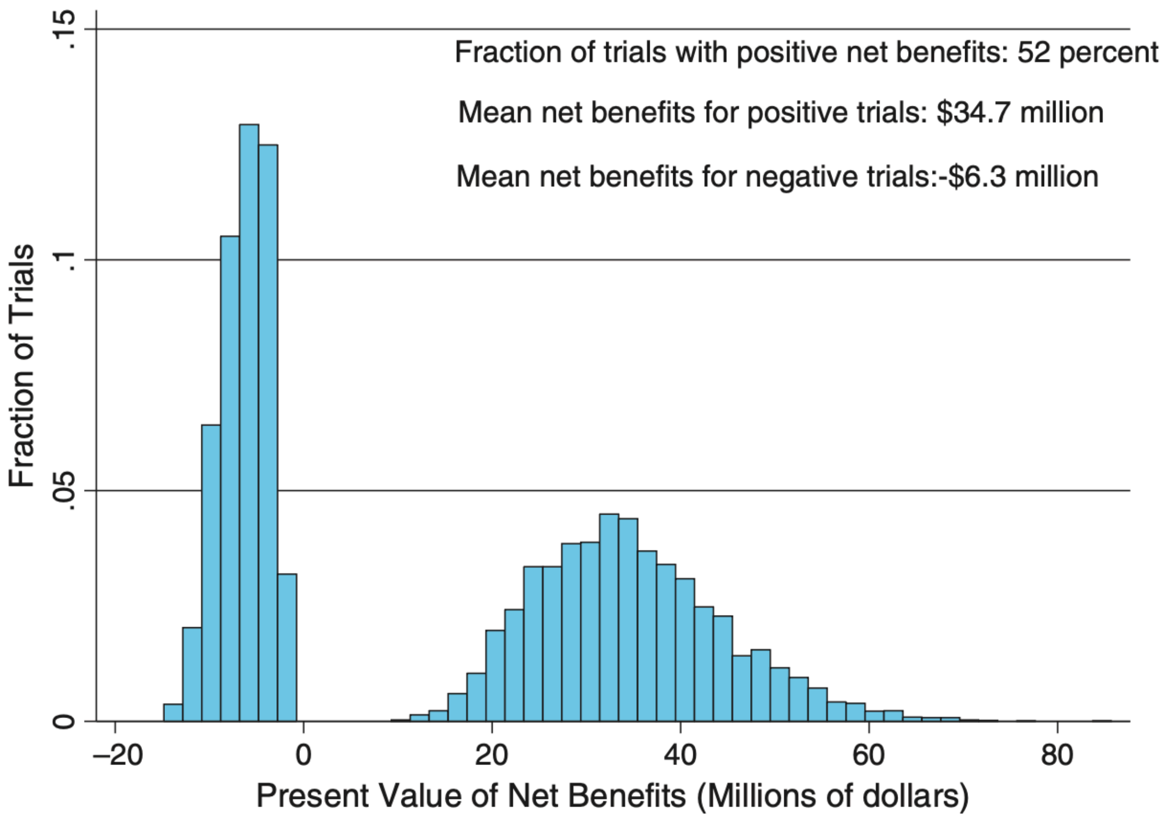
Monte Carlo simulation

- Enumerating all combinations is infeasible for many parameters → Then it is preferable to do a simulation analysis.

- Idea: specify probability distributions for uncertain inputs, draw many trials, compute net benefit each trial → empirical distribution of NB.

- Outputs to present: mean NB, standard deviation, histogram, P(NB > 0), and tail losses (expected shortfall).

- This is a histogram comprising 10,000 replications of random draws based on the bracketed assumptions listed in Table 11.2. The assumed distributions are all uniform, with the exception of hours lost *t*, which follows a normal distribution.



Monte Carlo simulation

- Select key uncertain parameters and plausible distributions (uniform if only bounds known; normal/triangular if you have center and spread).

- Run ≥1,000 trials (spreadsheet or statistical package). Record NB each trial.

- Report summary stats and visualize the histogram + probability of positive NB. Also test sensitivity to correlation assumptions if relevant.

- Use probability of positive NB to complement ENPV: even if mean ENPV > 0, a large downside tail may counsel staging or additional information.

Value of information

- **Idea:** Investing resources (time, money, effort) to reduce uncertainty is worthwhile if the information **increases the expected NPV** of the best action.

- **Measure:** value of information = increase in expected NPV from taking an optimal action *after* learning vs acting now.

$$\mathrm{VoI} = \mathbb{E}\left[\mathrm{ENPV}_{\text{after learning}}\right] - \mathrm{ENPV}_{\text{act now}}.$$

- **When it matters:** large irreversibility, big upside/downside asymmetry, and when learning is possible at low cost (surveys, pilots, delay to observe signals).

Setup (given):

- Irreversible investment $I_0 = 18{,}000{,}000$ NOK at $t = 0$.
- Annual operating profit is either $500{,}000$ or $1{,}500{,}000$ forever; each with probability 0.5.
- Discount rate $i = 0.04$.
- If we **delay 1 year**, we will learn the true profit with certainty and then invest only if profit = $1{,}500{,}000$.

Quasi-option value

Alternative 1 — Invest now (no learning):

- Expected annual profit $= 0.5(500{,}000) + 0.5(1{,}500{,}000) = 1{,}000{,}000.$
- PV of expected perpetuity $= 1{,}000{,}000/0.04 = 25{,}000{,}000.$
- $\text{ENPV}_{\text{now}} = 25{,}000{,}000 - 18{,}000{,}000 = 7{,}000{,}000.$

Alternative 2 — Wait 1 year (learn), then invest only if high profit:

- If high profit (prob 0.5): at $t = 1$ net value = $\frac{1{,}500{,}000}{0.04} - 18{,}000{,}000 = 19{,}500{,}000.$
- Discount that to $t = 0$: $19{,}500{,}000/(1.04) = 18{,}750{,}000.$
- Weighted by prob 0.5 → $\text{ENPV}_{\text{wait}} = 0.5 \times 18{,}750{,}000 = 9{,}375{,}000.$

Quasi-option / Value of waiting (VoI):

$$\text{VoI} = ENPV_{\text{wait}} - ENPV_{\text{now}} = 9{,}375{,}000 - 7{,}000{,}000 = 2{,}375{,}000 \text{ NOK}.$$

Implication: even though ENPV(now) is positive, the option value of waiting (learning) is larger — **postponement is optimal**.

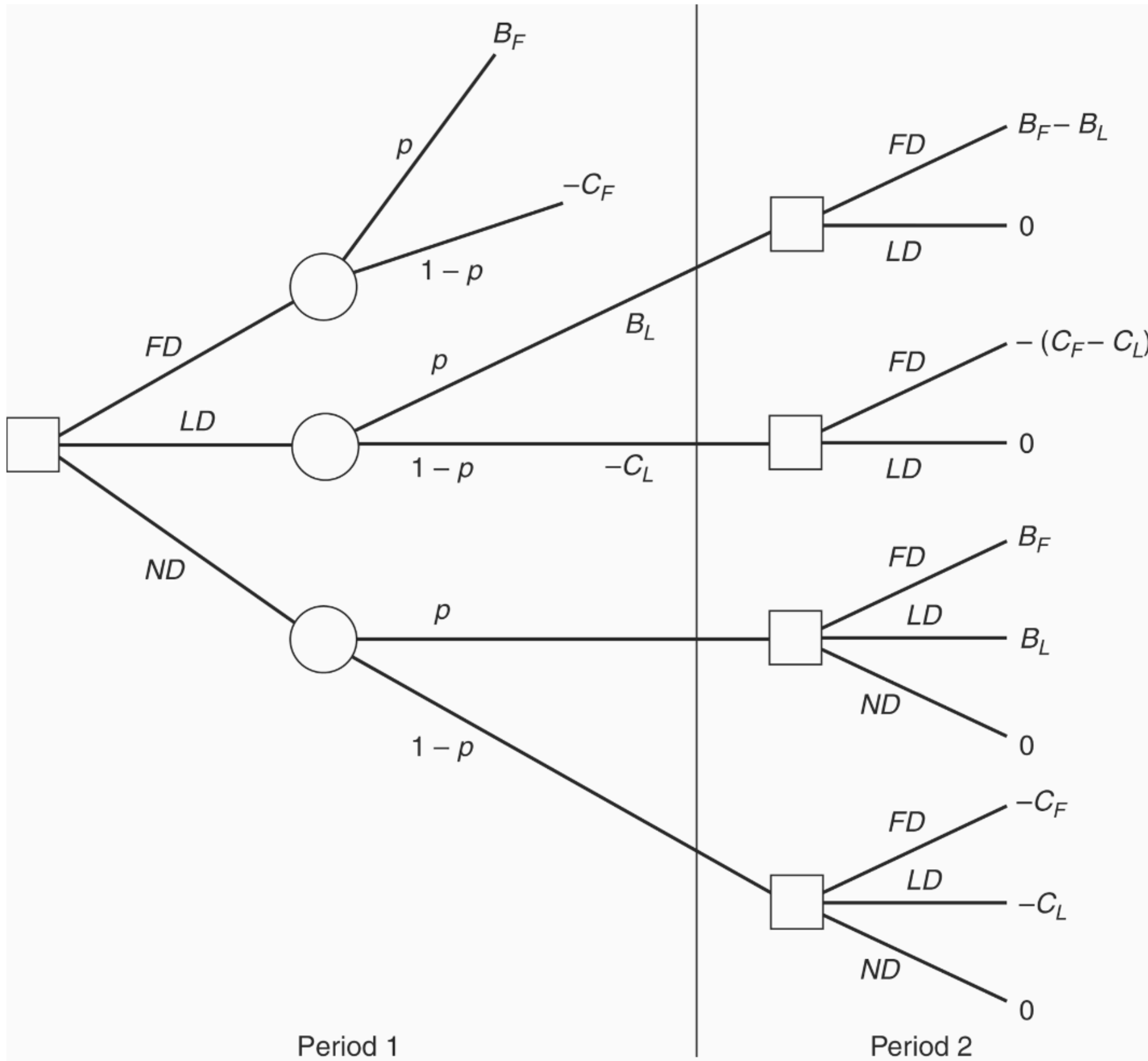
Quasi-option value

- Types of learning:
  —*Exogenous learning*: information arrives independently of our action (example above).
  —*Endogenous learning*: learning depends on actions. Build an explicit decision tree or dynamic model and use backward induction.

- Suppose government considers a development program with 3 choices in period 1:
  Full (FD, irreversible),
  Limited (LD, reversible / learnable), or
  No Development (ND = do nothing).

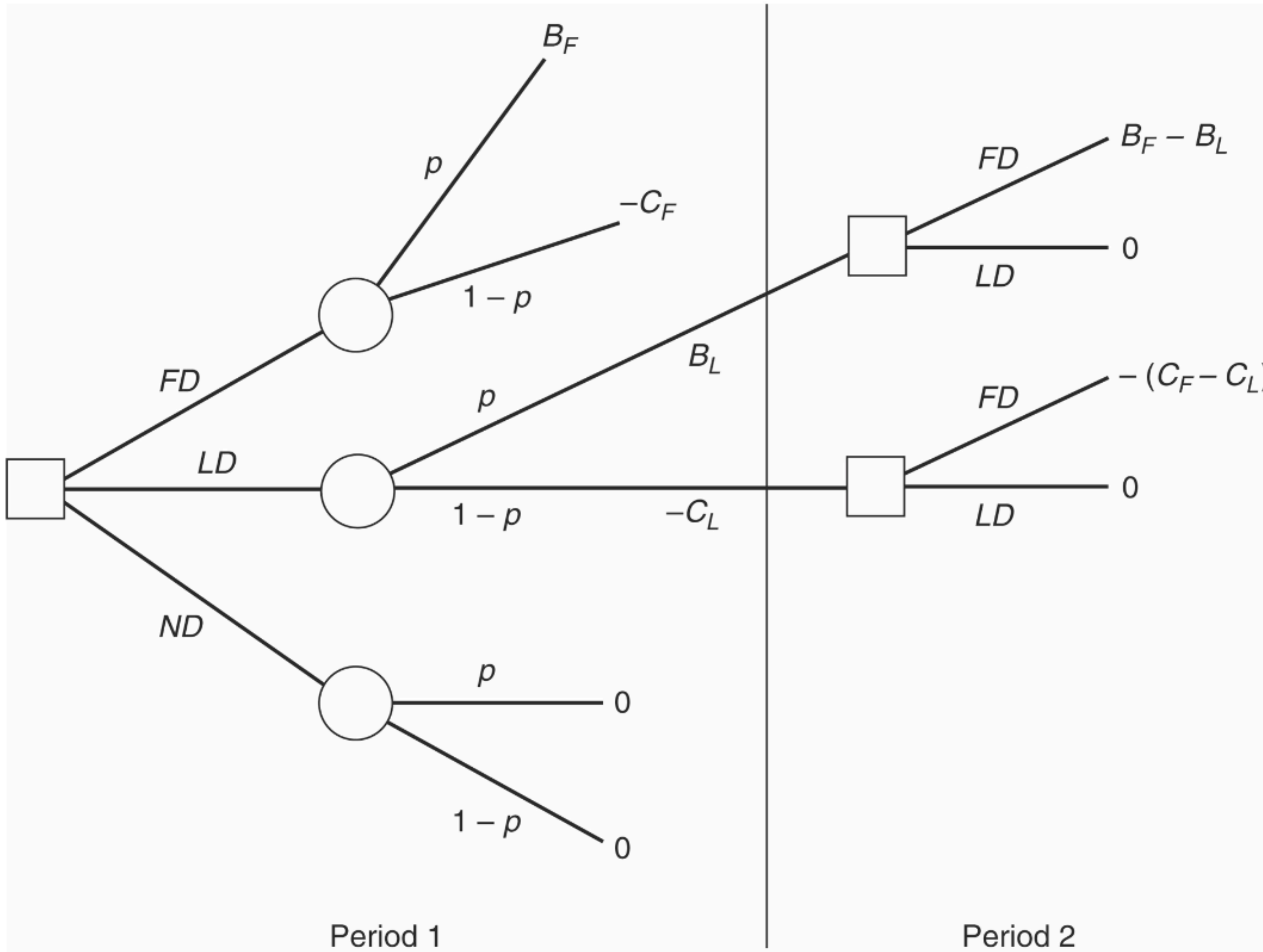
- Between periods we learn perfectly whether the project is **beneficial** or **costly** (two contingencies). Learning may be **exogenous** (occurs regardless of action) or **endogenous** (only generated by doing LD/FD).

Exogenous learning



	No learning	Exogenous learning	Endogenous learning
$E[FD]$	$pB_F - (1-p)C_F$	$pB_F - (1-p)C_F$	$pB_F - (1-p)C_F$
QOV		0	0
$E[LD]$	$pB_L - (1-p)C_L$	$p[B_L + (B_F + B_L)/(1+d)] - (1-p)C_L$	$p[B_L + (B_F - B_L)/(1+d)] - (1-p)C_L$
QOV		$p(B_F - B_L)/(1+d)$	$p(B_F - B_L)/(1+d)$
$E[ND]$	0	$pB_F(1+d)$	0
QOV		$pB_F/(1+d)$	$= 0$

Endogenous learning



	No learning	Exogenous learning	Endogenous learning
$E[FD]$	$pB_F - (1-p)C_F$	$pB_F - (1-p)C_F$	$pB_F - (1-p)C_F$
QOV		0	0
$E[LD]$	$pB_L - (1-p)C_L$	$p[B_L + (B_F + B_L)/(1+d)] - (1-p)C_L$	$p[B_L + (B_F - B_L)/(1+d)] - (1-p)C_L$
QOV		$p(B_F - B_L)/(1+d)$	$p(B_F - B_L)/(1+d)$
$E[ND]$	0	$pB_F(1+d)$	0
QOV		$pB_F/(1+d)$	$= 0$

A numerical example

$B_F = 100,\ B_L = 50,\ C_F = 80,\ C_L = 40,\ p(\text{benefit}) = 0.5,\ d = 0.8$

(Here $d = 0.8$ is the one-period discount factor; future payoffs are multiplied by d.)

Expected value	No learning	Exogenous learning	Endogenous learning
$E[FD]$	10.00	10.00	10.00
$E[LD]$	5.00	28.15	28.15
$E[ND]$	0.00	46.30	0.00

- **No learning:** immediate FD looks best.
- **Exogenous learning:** waiting (ND then FD if good) is best — **large value of information**.
- **Endogenous learning:** pursue limited program (LD) to learn, then scale up if good.
- Thus whether learning is exogenous or endogenous **dramatically changes** the optimal first-period action; explicitly model how information is generated before deciding.

Reading materials

- Boardman et al. Chapters 11.
- Veileder i samfunnsøkonomiske analyser, sections 3.4 and 3.6.